CHAPTER 1

SETS

CHAPTER ESSENCE AT A GLANCE:

Sets are just organized collections! Venn diagrams turn complicated word problems into visual maps. Master Union (all together), Intersection (shared only), and De Morgan's laws (flip union to intersection when you complement) and you score full marks!

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Roster vs Set-Builder:** Roster form lists explicit elements separated by commas {1, 2, 3}; Set-builder states defining rule {x : x is an integer and -3 < x < 7}.

• **Cardinality & Subsets:** A set with n elements has 2^n total subsets, 2^n - 1 proper subsets, and 2^n elements in its Power Set P(A). Empty set is a subset of EVERY set.

• **Set Operations:** Union (A B): elements in A or B or both. Intersection (A B): common elements in both A and B. Difference (A - B = A B'): elements in A but not in B.

• **De Morgan's Laws:** (1) (A B)' = A' B' (Complement of union is intersection of complements). (2) (A B)' = A' B'.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

Inclusion-Exclusion Principle for Two Sets

$$n(A \cup B \cup C) = \sum n(A) - \sum n(A \cap B) + n(A \cap B \cap C)$$

Inclusion-Exclusion Principle for Three Sets

$$(A \cup B)' = A' \cap B' \quad \text{and} \quad (A \cap B)' = A' \cup B'$$

De Morgan's Dual Laws of Set Complementation

$$n(\mathcal{P}(A)) = 2^{n(A)} \quad (\text{Proper Subsets} = 2^{n(A)} - 1)$$

Power Set and Proper Subsets Cardinality

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** The empty set is a subset of EVERY set, but is NOT an element of a set unless explicitly written inside { }!

• **Alert:** For word problems with two overlapping groups, draw a 2-circle Venn diagram and fill in the central intersection A B FIRST!

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Write all the subsets of the set A = {1, 2, 3}. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Identification & Formula]** Given set A = {1, 2, 3} has n = 3 elements. Total number of subsets is n(P(A)) = 2^n = 2^3 = 8. [1 Mark]

• **[Step 2: Evaluation]** Listing all 8 subsets: empty set (empty set), singletons {1}, {2}, {3}, pairs {1, 2}, {1, 3}, {2, 3}, and the set itself {1, 2, 3}. [1 Mark]

• **[Boxed Result: Subsets: empty set, {1}, {2}, {3}, {1, 2}, {1, 3}, {2, 3}, {1, 2, 3}]**

2. If A = {1, 2, 3, 4, 5, 6} and B = {2, 4, 6, 8}, find: (a) A union B (b) A intersection B (c) A - B (d) B - A. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Set Operations]** (a) A union B = {1, 2, 3, 4, 5, 6, 8}. [0.5 Mark]

• (b) A intersection B = {2, 4, 6}. [0.5 Mark]

• (c) A - B = Elements in A not in B = {1, 3, 5}. [0.5 Mark]

• (d) B - A = Elements in B not in A = {8}. [0.5 Mark]

• **[Boxed Result: A U B = {1,2,3,4,5,6,8}, A n B = {2,4,6}, A-B = {1,3,5}, B-A = {8}]**

3. State De Morgan's laws for any two sets A and B. Verify both laws using Venn diagrams. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Statement of De Morgan's Laws]** 1. (A union B)' = A' intersection B' (Complement of union is intersection of complements). 2. (A intersection B)' = A' union B' (Complement of intersection is union of complements). [1 Mark]

• **[Step 2: Venn Diagram Description for Law 1]** (A union B) is the region inside either circle. Its complement (A union B)' is the region outside both circles. A' is everything outside circle A, and B' is everything outside circle B. Their overlap A' intersection B' is exactly the region outside both circles. Regions match identically. [1 Mark]

• **[Step 3: Venn Diagram Description for Law 2]** (A intersection B) is the central lens. Its complement (A intersection B)' is everything except the central lens. A' union B' combines regions outside A and outside B, which also covers everything except the central overlap. Both laws verified. [1 Mark]

• **[Boxed Result: (A U B)' = A' n B' and (A n B)' = A' U B' verified]**

4. If U = {1, 2, 3, 4, 5, 6, 7, 8, 9}, A = {2, 4, 6, 8} and B = {2, 3, 5, 7}, verify that: (a) (A union B)' = A' intersection B' (b) (A intersection B)' = A' union B'. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Verification of Law (a)]** A union B = {2, 3, 4, 5, 6, 7, 8}. (A union B)' = U - (A union B) = {1, 9}. Now A' = {1, 3, 5, 7, 9} and B' = {1, 4, 6, 8, 9}. A' intersection B' = {1, 9}. LHS = RHS = {1, 9}. Law (a) verified. [1.5 Marks]

• **[Step 2: Verification of Law (b)]** A intersection B = {2}. (A intersection B)' = U - {2} = {1, 3, 4, 5, 6, 7, 8, 9}. A' union B' = {1, 3, 5, 7, 9} union {1, 4, 6, 8, 9} = {1, 3, 4, 5, 6, 7, 8, 9}. LHS = RHS. Law (b) verified. [1.5 Marks]

• **[Boxed Result: Both De Morgan's laws verified: (a) {1, 9}, (b) U - {2}]**

5. In a school there are 20 teachers who teach mathematics or physics. Of these, 12 teach mathematics and 4 teach both physics and mathematics. How many teach physics? [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Formula Setup]** Let M be the set of math teachers and P be physics teachers. Given: n(M union P) = 20, n(M) = 12, n(M intersection P) = 4. [1 Mark]

• **[Step 2: Union Formula]** n(M union P) = n(M) + n(P) - n(M intersection P). 20 = 12 + n(P) - 4. [1 Mark]

• **[Step 3: Solving for n(P)]** 20 = 8 + n(P) $\Rightarrow$ n(P) = 20 - 8 = 12. 12 teachers teach physics. [1 Mark]

• **[Boxed Result: Number of physics teachers = 12]**

6. In a survey of 60 people, it was found that 25 people read newspaper H, 26 read newspaper T, 26 read newspaper I, 9 read both H and I, 11 read both H and T, 8 read both T and I, and 3 read all three newspapers. Find: (a) The number of people who read at least one of the newspapers (b) The number of people who read exactly one newspaper. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: At Least One Newspaper (Union)]** Formula: n(H U T U I) = n(H) + n(T) + n(I) - n(H n T) - n(T n I) - n(H n I) + n(H n T n I). n(H U T U I) = 25 + 26 + 26 - 11 - 8 - 9 + 3 = 77 - 28 + 3 = 52. 52 people read at least one newspaper. [2 Marks]

• **[Step 2: Exactly One Newspaper]** Using 3-circle Venn diagram: H only = 25 - (8 + 3 + 6) = 25 - 17 = 8. T only = 26 - (8 + 3 + 5) = 26 - 16 = 10. I only = 26 - (6 + 3 + 5) = 26 - 14 = 12. Exactly one newspaper = 8 + 10 + 12 = 30 people. [2 Marks]

• **[Boxed Result: (a) At least one = 52 people, (b) Exactly one = 30 people]**

7. (a) If A and B are two sets such that n(A) = 3 and n(B) = 2. If (x, 1), (y, 2), (z, 1) are in A × B, find A and B, where x, y, and z are distinct elements. (2 Marks)n(b) Write all the proper subsets of set S = {a, b}. (2 Marks) [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Determining Sets A and B]** In Cartesian product A × B, the first coordinates belong to A and second coordinates belong to B. First elements: x, y, z ⇒ A = {x, y, z} (since n(A) = 3). Second elements: 1, 2 ⇒ B = {1, 2} (since n(B) = 2). [2 Marks]

• **[Step 2: Proper Subsets]** Proper subsets of a set S are all subsets of S EXCEPT the set S itself. S = {a, b} has total 2^2 = 4 subsets: empty set, {a}, {b}, {a, b}. Proper subsets = empty set, {a}, {b}. [2 Marks]

• **[Boxed Result: A = {x, y, z}, B = {1, 2}; Proper subsets = {empty set, {a}, {b}}]**